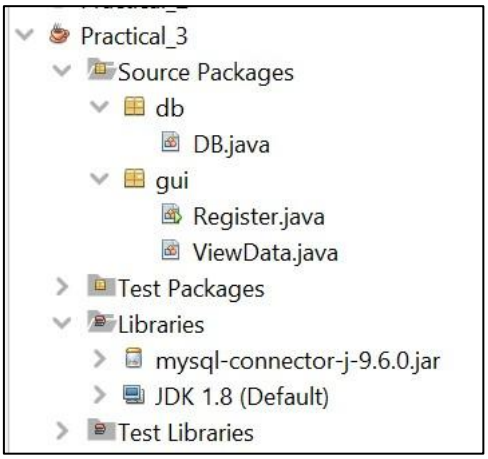
PRACTICAL 3

Data Driven GUI Application Develop an application using GUI for user registration. In first form user will enter the data for registration (e.g. Name, Birthdate, Email_id, Phone_No). When user clicks on submit button, data will be inserted into database and user will be redirected to another form. In this form data will be selected from database and displayed in proper format.



CODE:

DB.java

```
package db;

import java.sql.Connection;

import java.sql.DriverManager;

import java.sql.SQLException;


public class DB {

    public static Connection getCon() {

        Connection con = null;

        try {

            //Load Driver
```

```
            Class.forName("com.mysql.cj.jdbc.Driver");
            con = DriverManager.getConnection(
                "jdbc:mysql://localhost:3306/practical3_db",
                "root",
                ""
            );
            System.out.println("Database Connected Successfully");

        } catch (ClassNotFoundException e) {
            System.out.println("MySQL Driver Not Found");
        } catch (SQLException e) {
            System.out.println("Database Connection Failed");
        }
        return con;
    }
}
```

Register.java

```
package gui;
import db.DB;
import java.sql.Connection;
import java.sql.PreparedStatement;
import java.sql.SQLException;
import javax.swing.*;
import java.awt.event.ActionEvent;
import java.awt.event.ActionListener;
```

```
public class Register extends JFrame {
    JTextField txtName, txtEmail, txtPhone;
    JButton btnSave;
    public Register()
        setTitle("Registration Form");
        setSize(350, 250);
        setLayout(null);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);

        JLabel l1 = new JLabel("Name:");
        l1.setBounds(30, 30, 100, 25);
        add(l1);
        txtName = new JTextField();
        txtName.setBounds(120, 30, 150, 25);
        add(txtName);
        JLabel l2 = new JLabel("Email:");
        l2.setBounds(30, 70, 100, 25);
        add(l2);
        txtEmail = new JTextField();
        txtEmail.setBounds(120, 70, 150, 25);
        add(txtEmail);
        JLabel l3 = new JLabel("Phone:");
        l3.setBounds(30, 110, 100, 25);
        add(l3);
        txtPhone = new JTextField();
        txtPhone.setBounds(120, 110, 150, 25);
```

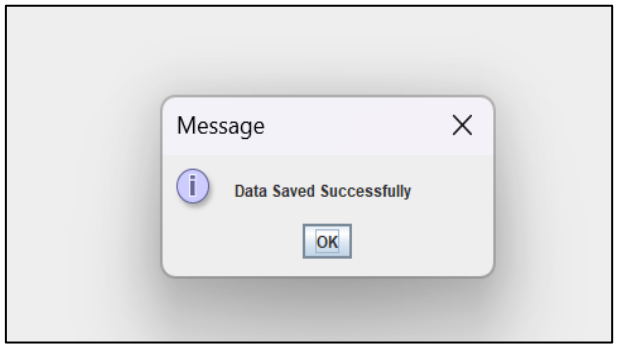
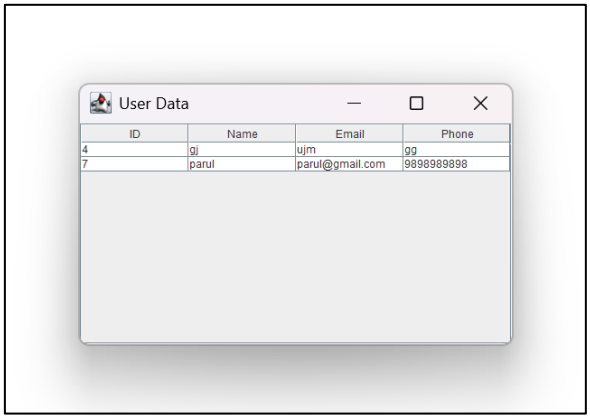
```
        add(txtPhone);
        btnSave = new JButton("Save");
        btnSave.setBounds(120, 160, 80, 30);
        add(btnSave);
        btnSave.addActionListener(new ActionListener() {
            public void actionPerformed(ActionEvent e) {
                saveData();
            }
        });
        setVisible(true);
    }
    void saveData() {
        Connection con = DB.getCon();
        if (con == null) {
            JOptionPane.showMessageDialog(this, "Database Not Connected");
            return;
        }
        try {
            String sql = "INSERT INTO user (name ,email_id ,phone_number )
VALUES(?,?,?)";
            PreparedStatement ps = con.prepareStatement(sql);
            ps.setString(1, txtName.getText());
            ps.setString(2, txtEmail.getText());
            ps.setString(3, txtPhone.getText());
            ps.executeUpdate();
            JOptionPane.showMessageDialog(this, "Data Saved Successfully");
            new ViewData();
```

```
            this.dispose();
        } catch (SQLException e) {
            JOptionPane.showMessageDialog(this, "Insert Operation Failed");
        }
    }
    public static void main(String[] args) {
        new Register();
    }}
```

ViewData.java

```
package gui;
import db.DB;
import java.sql.Connection;
import java.sql.ResultSet;
import java.sql.Statement;
import java.sql.SQLException;
import javax.swing.*;
import javax.swing.table.DefaultTableModel;

public class ViewData extends JFrame {
    JTable table;
    public ViewData() {
        setTitle("User Data");
        setSize(500, 300);
        setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        table = new JTable();
        DefaultTableModel model = new DefaultTableModel();
```

```
model.addColumn("ID");
model.addColumn("Name");
model.addColumn("Email");
model.addColumn("Phone");
table.setModel(model);
Connection con = DB.getCon();
if (con == null) {
    JOptionPane.showMessageDialog(this, "Database Not Connected");
    return;
}
try {
    Statement st = con.createStatement();
    ResultSet rs = st.executeQuery("SELECT * FROM `user`");

    while (rs.next()) {
        model.addRow(new Object[]{
            rs.getInt("id"),
            rs.getString("name"),
            rs.getString("email_id"),
            rs.getString("phone_number")
        });
    }

} catch (SQLException e) {
    JOptionPane.showMessageDialog(this, "Data Fetch Failed");
    e.printStackTrace(); // IMPORTANT
```

```
        }

        add(new JScrollPane(table));
        setVisible(true);
    }
}
```

OUTPUT: